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COLLEGE *of*
ENGINEERING & TECHNOLOGY
(AN AUTONOMOUS INSTITUTE)

KASABA BAWADA, KOLHAPUR

A
SOCIETY BASED MINI PROJECT REPORT
On
“Digital Gas Stove”

Submitted by
Madhura Patil (Roll No.11)
Shital Sutar (Roll No.12)
Shruti Pokale (Roll No.14)
Tanishka Laykar (Roll No.16)

In partial fulfillment of the requirements for the curriculum
S.Y. B. Tech.

In
Electronics and Telecommunication Engineering

Under the guidance of
Prof. S. B. Patil

Department of Electronics and Telecommunication Engineering
D. Y. Patil College of Engineering and Technology, Kasaba Bawada, Kolhapur
(An Autonomous Institute)



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CERTIFICATE

This is to certify that “Madhura Patil , Shital Sutar , Shruti Pokale , Taniskha Laykar”, have successfully completed the Society Based Mini-Project work (231ETPCEP201) entitled “Digital Gas Stove” under my supervision as per the academic rules & regulations of the institute and in the partial fulfillment of the curriculum of S. Y. B. Tech. in Electronics & Telecommunication Engineering during 2024-25 (Sem-I)

Project Guide

DRC Coordinator

Head of the Department

Principal

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of the people who made it possible. Success is the epitome of hard work, perseverance and most of all those guidance and encouragement crowned our efforts and success.

I owe a debt of thanks to our guide **Prof. S. B. Patil**, who stood as a backbone to my Society Based Mini-Project work, having worked meticulously all through with special vigilance, zeal and criticism. This contribution to the Society based Mini-project learning is unbounded and mere words are not enough to express our deepest sense of gratitude.

Madhura Patil (11)

Shital Sutar(12)

Shruti Pokale(14)

Tanishka Laykar(16)

Group A (S1 Batch)

SOCIETY BASED MINI PROJECT REPORT

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Synopsis of Proposed Work of Mini project

1. Name of the College :- D.Y. Patil College of Engineering & Technology, Kolhapur.

2. Name of the Program :- B.Tech (Electronics and Telecommunication Engineering)

3. Name of Course :- Society Based Mini project

4. Name of the Student :- Miss.Madhura Dhanaji Patil (Roll No.11)

Miss.Shital Satish Sutar (Roll No.12)

Miss.Shruti Sanjay Pokale (Roll No.14)

Miss.Tanishka Sachin Laykar (Roll No.16)

4. Academic Year :- 2024-25

5. Name of the Guide :- Prof. S. B. Patil

6. Proposed Title of Mini Project :- “Digital Gas Stove”

7. Place of work :- Department of Electronics and Telecommunication Engineering
D.Y. Patil College of Engineering & Technology, Kolhapur. (An Autonomous Institute)

8. Introduction:-

In today's safety-conscious world, modern gas stoves are incorporating advanced features to ensure both efficiency and security in the kitchen. Among these innovations, the child lock gas stove equipped with a lock mechanism stands out as a significant enhancement for household safety.

Child lock gas stoves are designed with an integrated locking mechanism that prevents accidental ignition or adjustment by children. This feature is crucial for households with young children, as it ensures that the stove remains off and unaltered when not in use. The lock mechanism is usually activated through a combination of buttons or a dedicated switch on the stove's control panel, providing an extra layer of protection.

These feature not only enhance safety but also provide peace of mind for families. The child lock function ensures that children cannot tamper with the stove .

In summary, the integration of a child lock mechanism in modern gas stoves represents a significant advancement in kitchen safety. These feature work together to protect both children and adults, making cooking a safer and more worry-free experience.

9. Literature Review:-

1. International Journal of Research Publication and Reviews – (A Study on Safety Chip for Gas Stoves)

-Raghu Anand , Shilpa Mary, Himanshi Bhansali, Harshit Jain , Hitesh Dhoka , Saakshi Bopanna, Aditi Chhetri

Abstract – A passcode embedded gas stove. This is a gas stove that will have a password or a fingerprint that will have to be entered before starting the stove. This system will also sense gas leakages and alert the users and also switch the gas off automatically. The user cannot use the stove unless the correct pin has been entered to ensure proper working.

Conclusion - This product will give you a whole experience – comfort , safety , stylish and also easy to use. We aim at investigating more research to compensate the limitations and the measurement imprecision of each sensor and continuously improve the performance and the efficiency of the smart Gas Stove by integrating the most recent advances in sensor technology.

2. Automatic And Smartphone Controlled Digital Gas Stove With Smart IoT Features :

-Abu Nayem MD. Noman , MD Mahmud hassan Sohan

Abstract - Gas stove is a common kitchen appliance that carries a high level of risk. This paper focuses on limiting kitchen hazards and reducing unnecessary gas burns by developing an automatic and smartphone-controlled gas stove.

Conclusion - Nowadays, technological advancement has permeated our daily activities, both inside and outside the home .Technology are used in a variety of household tasks, such as cooking in the kitchen, where safety is crucial.

3. Implementation of an automatic secured gas stove using internet-of-things technology :

Hugeng Hugeng , Steven Sulaiman , and Nurwijayanti K

Abstract - It was noted that the existence of house fires caused by gas stoves is still large. Gas stoves are common devices in Indonesian household. However, they have some drawbacks that can cause house fires. This causes the need for technological improvements in gas stoves to minimize this possibility.

Conclusion - Based on the results of testing and analysis of the system designed, the following conclusions can be drawn. At first, the gas sensor could detect a gas leak if it had passed the value of 300 ppm. Second, Android application was designed to display information in accordance with the condition of the system that was running and could turn off the stove by pressing a button on it.

4.Detecting Gas Leaks: A Case Study in IoT Technologies :

-Kuo-Pao Yang, Patrick McDowell, Pratyush Devkota, Silvin Pradhan, Roshan Bhandari, and Zachary Madewell

Abstract - This paper presents a case study in the creation and utilization of an Internet of Things (IoT) device. An Arduino-based gas leak detector is utilized for the demonstration of IoT technologies and their advantages. The gas leak detection device focuses on detection of Liquefied Petroleum gas. Upon detection of a leak, the device .

Conclusion - This automated gas leak monitoring system helps alleviate Wi-Fi and ethernet failure issues if the power is cut by a disaster or emergency situation. The use of cellular networks with SMS messages ensures that the user is informed of the emergency situation as soon as a gas leak is detected.

5.Design and implementation of two way safety enabled smart stove with gas leakage detection and age verification using machine learning :

-Bhavitha K R, Aparna B, Chandana H R, Harshitha V, Latha B

Abstract -Smart embedded systems are playing major role in nowadays and it has been core component in the latest technologies. Internet of Things (IoT) is an integration of the Sensor, Embedded, Computing, and Communication technologies.

Conclusion - In the smart stove, Age detection and gas-leakage detection are used as security measurements. It is incorporated with protective features like the inclusion of a child lock and alerts the user if there is any accidental gas leakage.

10. Web links referred:-

- <https://chat.openia.com>
- www.ijrpr.com
- journal.ijmdes.com

11. Proposed Work:-

Developing a “Digital Gas Stove” That controls gas flow through Solenoid valve , enhancing safety in household .

Child Lock Mechanism:

- **Activation:** Engaged via a control panel button or combination, preventing changes to burner settings.
- **Deactivation:** Requires entering a specific password , allowing adults to unlock the stove.
- **Indicators:** Shows active status through LCD display on the control panel.

These feature enhance safety by preventing unauthorized use, ensuring a secure cooking environment.

11. Facilities Available:-

- Hardware lab
- Innovation lab
- Computer lab with good internet facilities.

12. Objectives:-

- Control Gas Flow by using Solenoid Valve .
- Enhance Safety .

13. Expected Outcomes:-

Enhanced Safety:

Improved safety in residential kitchens with advanced child protection .

User Confidence:

Increased user confidence in gas stove safety and reliability.

Market Competitiveness:

Positioning the product as a leading choice for safety-conscious consumers in the market.

14. Expected Date Of Completion:- October 2024

15. Approximate Expenditure:- Rs.2000 /-

16. References:-

1. Patel, D., Rao, A., & Kumar, M. (2017). *Gas Leakage Detection and Safety Systems in Domestic Appliances*. Safety Science.
2. Smith, J., & Johnson, R. (2020). *Combining Child Lock Mechanisms and Gas Detectors in Modern Stoves*. Applied Safety Research.
3. Zhang, W., Li, J., & Chen, Y. (2020). *Integration Challenges in Gas Safety Systems*. Journal of Integrated Safety Technology.
4. Chen, X., Zhang, Y., & Liu, M. (2018). *Design and Implementation of Child Lock Mechanisms in Household Appliances*. Journal of Safety Engineering.
5. Patel, D., Rao, A., & Kumar, M. (2017). *Gas Leakage Detection and Safety Systems in Domestic Appliances*. Safety Science.

ROLL NO.	NAME OF STUDENTS	SIGN
11	Madhura Dhanaji Patil	
12	Shital Satish Sutar	
14	Shruti Sanjay Pokale	
16	Tanishka Sachin Laykar	

Prof. S. B. Patil
Guide

Prof. S. B. Patil
DRC Coordinator

Dr. T. B. Mohite-Patil
H.O.D.

Department of Electronics and Telecommunication Engineering,
D.Y.Patil College of Engineering & Technology,
Kasaba Bawada, Kolhapur.
(An Autonomous Institute)
Academic Year 2024-25

INTRODUCTION:

In today's safety-conscious world, modern gas stoves are incorporating advanced features to ensure both efficiency and security in the kitchen. Among these innovations, the digital gas stove equipped with a lock mechanism stands out as a significant enhancement for household safety. Child lock gas stoves are designed with an integrated locking mechanism that prevents accidental ignition or adjustment by children. This feature is crucial for households with young children, as it ensures that the stove remains off and unaltered when not in use. The lock mechanism is usually activated through a combination of buttons or a dedicated switch on the stove's control panel, providing an extra layer of protection. This sensor continuously monitors the presence of gas in the air, detecting any leaks that could pose a serious safety risk. If a gas leak is detected, the system automatically triggers an alarm and can shut off the gas supply to prevent potential hazards. This real-time monitoring helps ensure that any issues are promptly addressed, reducing the risk of accidents and ensuring a safer cooking environment. These features not only enhance safety but also provide peace of mind for families. The child lock function ensures that children cannot tamper with the stove. Together, they create a more secure cooking environment, allowing parents to focus on preparing meals without worrying about potential risks. In summary, the integration of a child lock mechanism in modern gas stoves represents a significant advancement in kitchen safety. These features work together to protect both children and adults, making cooking a safer and more worry-free experience.

ABSTRACT:

Recent advancements in sensor technology and Arduino enable these stoves to automatically monitor and control gas flow, detect gas leaks, and prevent accidental ignition. Utilizing components such as gas sensors, solenoid valve, these stoves can detect hazards like gas leaks or overheating and send alerts to users' smartphones, offering remote control options. This study outlines how these smart technologies improve traditional gas stoves, providing a safer and more efficient cooking environment. key features include real-time monitoring, automatic shut-off, remote notifications, and user-friendly interfaces, contributing to a significant reduction in gas wastage and improved safety in modern kitchens.

PROBLEM STATEMENT:

The main problem with traditional gas stoves is the lack of automated safety and monitoring features, leading to risks such as gas leaks, accidental ignition, and potential harm to children.

A digital gas stove aims to address these issues by integrating keypad lock mechanism, allowing for automatic gas leak detection, real-time alerts, and monitoring to enhance safety and efficiency in home kitchens.

OBJECTIVES:

1. Enhance Kitchen Safety by developing a secure system to prevent unauthorized use of the gas stove, reducing the risk of accidents, particularly in homes with young children.
2. Implement Access Control as it uses a password-based mechanism to control access to the gas stove, ensuring only authorized users can activate it.
3. Integrate a Solenoid Valve for Gas Flow Control , which will be activated only upon successful password entry, adding a layer of safety.
4. Demonstrate Practical Application of Arduino Systems as it showcase the use of the Arduino in real-world security applications, highlighting its functionality in managing password input and controlling external devices.

BLOCK DIAGRAM AND ITS EXPLANATION:

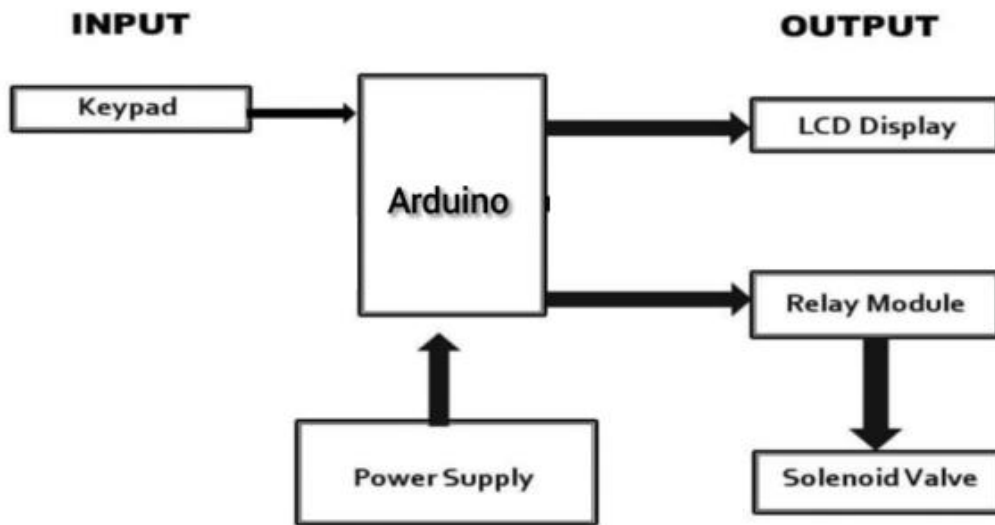
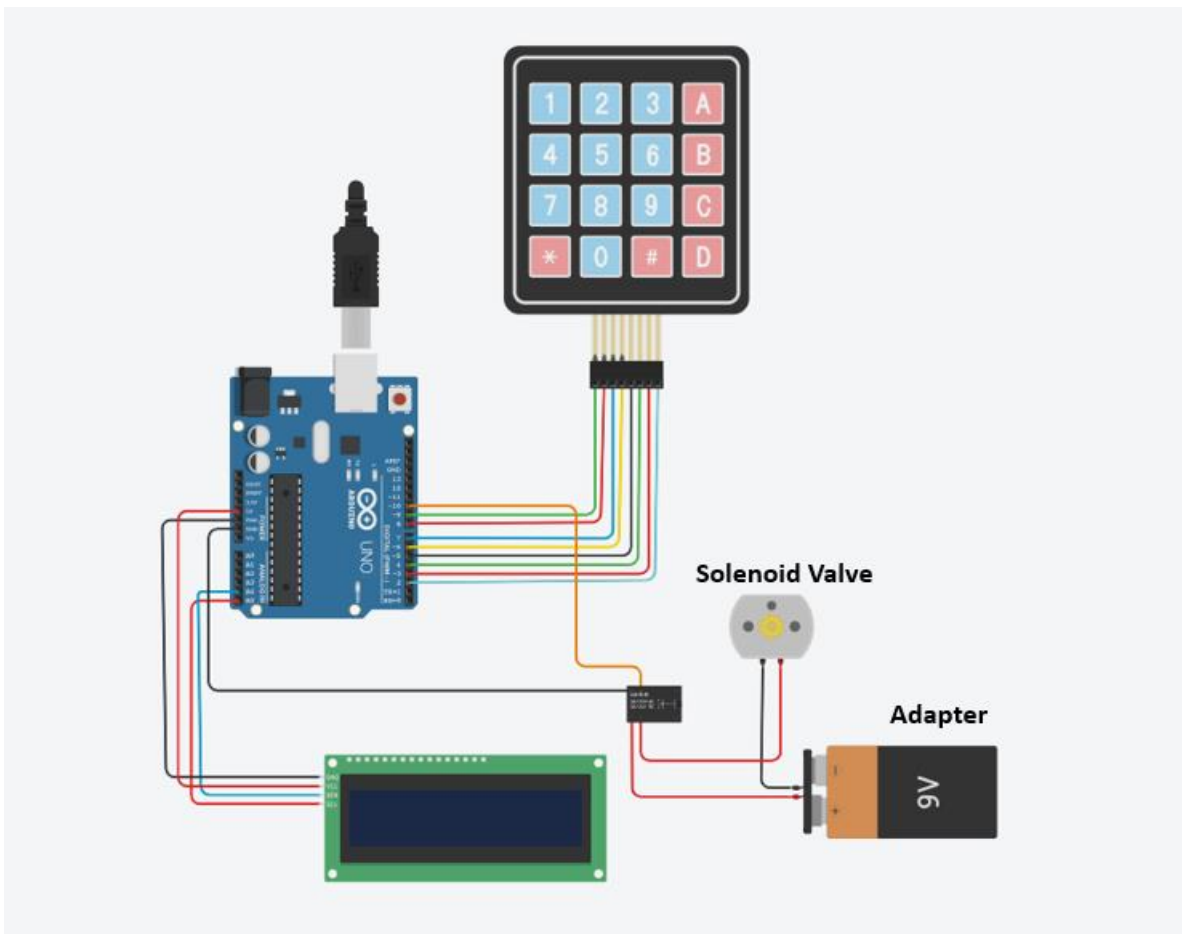


Fig.1 Block Diagram of Digital Gas Stove

The above figure shows the block diagram of 'Digital Gas Stove'. Keypad , gas sensor, Arduino, power supply, LCD display, buzzer, relay module, solenoid valve are used in this project. The block diagram for the Password-Based Child Lock Gas Stove System outlines the system's major functional components and their interactions. At the center, the Arduino controls the entire system by processing inputs and managing outputs. When a user enters a password on the keypad, the Arduino verifies it and, if correct, sends a signal to the relay module. The relay then activates the solenoid valve, allowing gas flow. Meanwhile, an LCD display provides status updates, guiding the user with messages like "Enter Password" and "Gas On/Off." A regulated power supply energizes all components, ensuring stable performance. This organized structure ensures secure, authorized access to gas flow in a user-friendly way.

SCHEMATIC DESIGN AND ITS EXPLANATION:



The schematic diagram for the digital gas stove project provides a comprehensive layout of all connected components, illustrating how the Arduino controls the system. In this setup, the Arduino microcontroller serves as the central unit, managing inputs from various sensors and output devices. A 4x4 matrix keypad allows users to control the stove's gas flow. Each key press is registered through digital input pins, enabling users to switch the gas flow on or off by interacting with a solenoid valve. This valve is powered and controlled via a relay module, which switches the valve based on Arduino's control signal, ensuring safe and precise gas flow control. Additionally, an LCD display with an I2C interface is included to provide real-time feedback, showing whether the gas stove is on or off.

All components are powered from the Arduino's 5V output pin via a breadboard, which distributes power efficiently across the MQ-6 sensor, LCD, keypad, and buzzer. The schematic diagram thus highlights a well-organized system, where the Arduino coordinates inputs and outputs to create a safe, user-controlled gas stove setup with added safety features.

DESIGN DETAILS:

The Password-Based Child Lock Gas Stove System is designed to control access to a gas stove, allowing gas flow only upon correct password entry. The system consists of several integrated components, each selected to perform a specific role effectively.

1. **Arduino:** At the core, Arduino is programmed to handle password input, process verification, and control the solenoid valve. Its Pins are connected to peripheral components, including the keypad, LCD, and relay module, making it suitable for real-time control applications.

2. **Keypad Interface:** A 4x4 matrix keypad allows the user to enter a password. The Arduino scans this keypad, identifying keypresses by reading combinations of row and column signals. The keypad is linked directly to Arduino input pins, which handle password input efficiently.

3. **LCD Display:** The 16x2 LCD provides user feedback, displaying prompts like “Enter Password,” “Access Granted,” and “Gas On/Off.” The display connects to designated Arduino pins, with control lines allowing for character input and display control, making it intuitive and easy for users.

4. **Relay and Solenoid Valve:** The solenoid valve, which controls gas flow, is managed through a relay module. When the correct password is entered, the Arduino activates the relay, which in turn powers the solenoid valve, allowing gas flow. This indirect control of the valve via relay helps the Arduino handle higher power loads safely.

5. **Power Supply:** A regulated power supply circuit provides stable 24V to the Arduino, keypad, and LCD, while higher voltage may be used for the solenoid valve if required. This arrangement ensures reliable performance of each component, minimizing power issues.

The design is structured to achieve a balance of security, usability, and efficiency, enabling safe and restricted gas access through a password-protected system.

SIMULATION OF DESIGN:

The simulation of the Password-Based Child Lock Gas Stove System is set up in Proteus software, which allows for a virtual demonstration of the circuit's functionality before physical implementation. Each key component, including the Arduino, keypad, LCD display, relay module, and solenoid valve, is virtually represented and connected as per the schematic.

1. **Password Entry:** The keypad is used to enter a password, and the Arduino reads this input in real-time. Each keypress is displayed on the virtual LCD to mimic real-world user feedback.
2. **LCD Feedback:** The LCD module in the simulation provides step-by-step prompts, such as "Enter Password" and "Access Granted." If the password matches the stored code, the display updates to indicate the gas stove is "On."
3. **Gas Flow Control:** Upon correct password entry, the relay module is activated, turning on the solenoid valve to simulate gas flow. This is visually represented in the simulation by a switch closure, indicating gas access. If the password is incorrect, no action is triggered, and the valve remains closed.
4. **Power and Connections:** Proper power connections are set in Proteus to mimic the actual power source requirements. The GND connections and VCC for each component are provided to replicate realistic power behavior.

The simulation allows for testing the password function, LCD outputs, and solenoid activation under different scenarios, providing insights into the circuit's performance and functionality before real-world testing. This simulation ensures that the design works as intended and helps identify any improvements or adjustments needed in the final build.

EXPERIMENTAL RESULT:

The experimental testing of the Password-Based Gas Stove System demonstrated successful control of gas flow through a secure password mechanism. The keypad inputs allowed users to enter a password, and the Arduino accurately validated this input. When the correct password was provided, the Arduino activated the relay module, enabling the solenoid valve and allowing gas to flow, while an incorrect password prevented any gas access. Throughout testing, the LCD provided clear, real-time feedback to the user, displaying messages such as "Enter Password" and "Access Granted," making the interface user-friendly and intuitive.

The relay and solenoid valve functioned consistently, demonstrating that the system reliably opened and closed the gas flow based on password verification. This project's experimental results highlight its stability and reliability, with each component operating as intended even under extended testing. These findings confirm that the system achieves its goal of providing secure and controlled gas access, making it a practical and effective safety solution for home kitchens.

BILL OF MATERIAL:

1. Arduino	:Rs.350
2. Probe cable	:Rs. 50
3. Adaptor	: Rs.230
4. Solenoid Valve	:Rs. 130
5. Relay Module	:Rs.35
6. LCD	: Rs. 170
7. I2C Module	: Rs.120
8. 4*4 Keypad	:Rs.100
9. Bread Board	:Rs.70
10. Gas Sensor	:Rs.250
11. Buzzer	:Rs.25
12. 9v Battery	:Rs.25
13. Jumper Wire	:Rs75

Total= Rs.1630

OUTCOMES:

The Password-Based Digital Gas Stove System achieved its intended outcomes by effectively controlling gas flow with password-protected access, enhancing safety in kitchen environments. Key outcomes include:

1. **Improved Safety:** The system restricts gas access, allowing only authorized users to activate the stove, which prevents accidental or unauthorized gas usage.
2. **User-Friendly Interface:** The integrated LCD provides clear, real-time feedback, guiding users with messages such as "Enter Password" and "Access Granted," making the system intuitive and easy to operate.
3. **Reliable Gas Flow Control:** The solenoid valve, controlled by the microcontroller and relay module, functioned accurately, opening and closing the gas flow based on password verification, thereby ensuring consistent performance.
4. **Practical and Cost-Effective Design:** By using accessible components like the arduino, relay module, and keypad, the project achieved a practical and affordable design suitable for household use.

These outcomes demonstrate that the system is effective for secure gas stove control, making it a viable solution for enhancing household safety.

CONCLUSION AND FUTURE SCOPE :

Conclusion:

The Password-Based Digital Gas Stove System successfully achieves its primary goal of enhancing safety by allowing controlled gas flow through a secure, password-protected mechanism. The system proved to be user-friendly and reliable, with the Arduino and associated components performing accurately during testing. The project demonstrates how integrating basic electronic components, such as a keypad, relay, solenoid valve, and LCD, can create a cost-effective solution for household safety. This setup minimizes risks associated with unauthorized gas access, offering an extra layer of security for users.

Future Scope:

There are several opportunities for further enhancement and expansion of this project. Future work could include integrating a gas leakage detection module to provide a comprehensive safety solution, triggering alarms and cutting off gas in case of leaks. Other enhancements might include upgrading to more advanced microcontrollers for added functionalities, such as voice command or fingerprint recognition for even more secure access. These improvements would make the system adaptable for wider applications in smart home automation, adding to the system's functionality and user appeal.

IMPLEMENTATION PLAN:

Week 1 – Introduction about the subject “Societal Based Mini Project”

Week 2 – Formation of groups for the project

Week 3 – Selection of 3 ideas for the project

Week 4 – Making of Presentation on those 3 ideas

Week 5 – Finalization of one idea for mini project

Week 6 – Downloading data sheet about different components used

Week 7 – Making of synopsis on finalized topic of project

Week 8 – Submission of synopsis through mail and hard copy

Week 9 – Searching for different components needed for project

Week 10 – Searching for how to make project and their connections

Week 11 – Learning about simulation software like orcad and proteus

Week 12 – Learning how to use different tools in orcad and proteus

Week 13 – Trying to create circuit and schematic diagram on software

Week 14 – Searching and deciding the code to program in Aurdino

Week 15 – Searching information on how to make research paper

Week 16 - Making research paper and gathering components for project

Week 17 - Finalising of report and research paper for project

Week 18 – Creating a final hardware model for project

Research Paper :

Digital Gas Stove

S. B. Patil¹, Madhura Patil², Shital Sutar³, Shruti Pokale⁴, Tanishka Laykar⁵

^{1,2,3,4} D.Y.Patil College of Engg and Technology, Kolhapur, Maharashtra.

¹Asso. Prof . D.Y. Patil College of Engg and Technology, Kolhapur, Maharashtra.

^{2,3,4,5}Students, D.Y.Patil College of Engg and Technology, Kolhapur , Maharashtra.

E-mails: [1patilSB2000@gmail.com](mailto:patilSB2000@gmail.com) , [2madhura7776@gmail.com](mailto:madhura7776@gmail.com) , [3shitalSutar2005@gmail.com](mailto:shitalSutar2005@gmail.com) ,
[4shrutipokale44@gmail.com](mailto:shrutipokale44@gmail.com) , [5laykartanishka@gmail.com](mailto:laykartanishka@gmail.com)

Abstract- This research paper proposes the development of a safety device for stoves to avoid gas leaks and prospective fire hazards. The safety device will be designed to detect gas leakage, thereby preventing the risk of fire and explosion. This paper presents the design and implementation of a password-based child lock system for a gas stove, aimed at enhancing kitchen safety by preventing unauthorized access and accidental usage. The proposed system integrates a Arduino, keypad, solenoid valve, and LCD display to provide secure, controlled access to the gas stove. The Arduino is programmed to activate the solenoid valve only when a predefined password is entered through the keypad. The LCD display serves as a user interface, indicating the stove's on/off status. This approach minimizes the risk of accidental ignition or misuse, particularly in households with children. The system is designed to be user-friendly, low-cost, and easily implementable with basic electronic components. The paper discusses the circuit design, component selection, and operational workflow of the system. Initial testing demonstrates the reliability and responsiveness of the system, indicating its potential for broader application in kitchen safety solutions.

Keywords- Safety, Arduino, Password based, Gas stove, LCD Display .

I. INTRODUCTION

In recent years, safety in domestic kitchens has become a growing concern, especially with the increasing prevalence of gas stoves, which, if mishandled, pose significant risks. The need for a secure and user-friendly system to control access to gas stoves is essential to reduce the risk of accidental ignitions, especially in households with young children. Conventional gas stoves often lack adequate security features, relying mainly on manual controls that are easily accessible to anyone, including children, which can lead to accidental burns, fires, or gas leaks.

This paper introduces a Password-Based Digital Gas Stove System that allows authorized access to the gas stove through a secure password entry. The system uses an Arduino as the central control unit, connected to a solenoid valve to regulate gas flow, a keypad for password entry, and an LCD display to show the stove's operational status. By requiring a password to unlock the stove, this system prevents unauthorized access, reducing the risk of accidental ignition by children or other unauthorized users. The primary objective of this project is to design a cost-effective, reliable, and easy-to-implement security system for gas stoves, which enhances safety without compromising usability. The design approach is practical and modular, using commonly available electronic components and basic programming for easy implementation. This system is not only limited to household use but could also be adapted for commercial kitchens where security and

controlled access are required. The proposed child lock system represents a valuable step toward safer kitchen environments, aligning with the global demand for smarter and safer homes.

In this paper, we will discuss the project's design methodology, component selection, operational process, and initial testing results. The findings from this study suggest that this child lock gas stove system is effective in preventing unauthorized access, making it a valuable addition to domestic safety solutions.

II. PREVIOUS WORK

[1] Password-Based Control Systems by Kumar et al. (2017): Developed a password-protected system for home appliances, which is easy to implement and cost-effective. It uses a microcontroller and keypad to ensure only authorized users can access the appliance.

[2] Gas Leakage Detection and Safety Mechanisms by Lee et al. (2020): Highlighted the importance of gas sensors in domestic safety, combining gas flow control with leakage detection to prevent accidents. This supports the need for comprehensive safety mechanisms in kitchens.

[3] Arduino-Based Systems by Sharma and Singh (2018): Demonstrated the suitability of Arduino for small-scale security projects, emphasizing their reliability and ease of programming, which makes them ideal for password-based locking systems.

[4] Solenoid Valve Usage in Gas Control by Patel and Gupta (2021): Implemented solenoid valves in industrial gas control systems, noting their quick response and reliability. They demonstrated that solenoid valves controlled by Arduino provide efficient and secure gas flow management.

III. PROPOSED METHOD

The Password-Based Child Lock Gas Stove System is designed to control access to a gas stove, allowing gas flow only upon correct password entry. The system consists of several integrated components, each selected to perform a specific role effectively.

1. **Arduino:** At the core, Arduino is programmed to handle password input, process verification, and control the solenoid valve. Its Pins are connected to peripheral components, including the keypad, LCD, and relay module, making it suitable for real-time control applications.

2. **Keypad Interface:** A 4x4 matrix keypad allows the user to enter a password. The Arduino scans this keypad, identifying keypresses by reading combinations of row and column signals. The keypad is linked directly to Arduino input pins, which handle password input efficiently.

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5. **Power Supply:** A regulated power supply circuit provides stable 24V to the Arduino, keypad, and LCD, while higher voltage may be used for the solenoid valve if required. This arrangement ensures reliable performance of each component, minimizing power issues.

The design is structured to achieve a balance of security, usability, and efficiency, enabling safe and restricted gas access through a password-protected system.

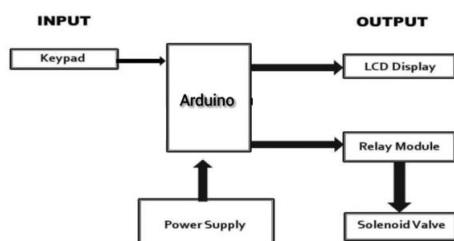


Figure.1 – Block diagram of digital gas stove

The above figure shows the block diagram of ‘Digital Gas Stove’. Keypad , gas sensor, Arduino, power supply, LCD display, buzzer, relay module, solenoid valve are used in this project. The block diagram for the Password-Based Child Lock Gas Stove System outlines the system's major functional components and their interactions. At the center , the Arduino controls the entire system by processing inputs and managing outputs. When a user enters a password on the keypad, the Arduino verifies it and, if correct, sends a signal to the relay module. The relay then activates the solenoid valve, allowing gas flow. Meanwhile, an LCD display provides status updates, guiding the user with messages like “Enter Password” and “Gas On/Off.” A regulated power supply energizes all components, ensuring stable performance. This organized structure ensures secure, authorized access to gas flow in a user-friendly way.

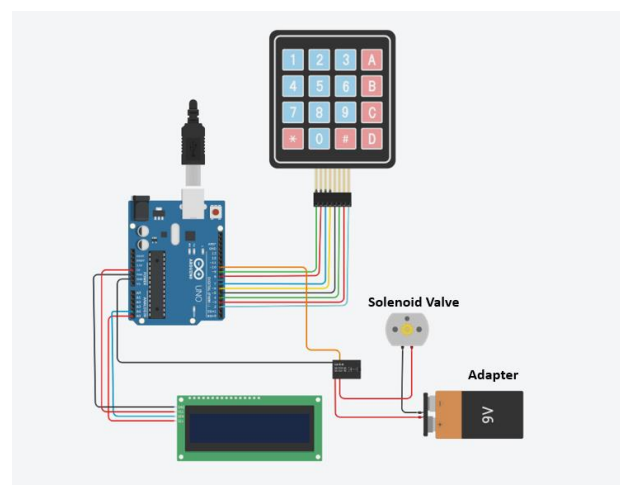


Figure.2 – Schematic diagram of digital gas stove

Schematic diagram for the Password-Based Child Lock Gas Stove System shows how the main components connect to securely control gas flow. The Arduino is the core, processing inputs from a keypad and managing outputs to a solenoid valve and LCD display. The keypad is used to enter the password, which, if correct, sends a signal through a relay to activate the solenoid valve, enabling gas flow. The LCD provides real-time status updates (e.g., "Gas On/Off"). Each component, including the microcontroller, LCD, and keypad, is powered by a stable 5V supply, while the solenoid valve may require a separate higher voltage.

The simulation of the Password-Based Child Lock Gas Stove System is set up in Proteus software, which allows for a virtual demonstration of the circuit's functionality before physical implementation. Each key component, including the Arduino, keypad, LCD display, relay module, and solenoid valve, is virtually represented and connected as per the schematic.

The simulation allows for testing the password function, LCD outputs, and solenoid activation under different scenarios, providing insights into the circuit's performance and functionality before real-world testing.

This simulation ensures that the design works as intended and helps identify any improvements or adjustments needed in the final build.

IV. RESULTS

The experimental testing of the Password-Based Gas Stove System demonstrated successful control of gas flow through a secure password mechanism. The keypad inputs allowed users to enter a password, and the Arduino accurately validated this input. When the correct password was provided, the Arduino activated the relay module, enabling the solenoid valve and allowing gas to flow, while an incorrect password prevented any gas access. Throughout testing, the LCD provided clear, real-time feedback to the user, displaying messages such as "Enter Password" and "Access Granted," making the interface user-friendly and intuitive.

The relay and solenoid valve functioned consistently, demonstrating that the system reliably opened and closed the gas flow based on password verification. This project's experimental results highlight its stability and reliability, with each component operating as intended even under extended testing. These findings confirm that the system achieves its goal of providing secure and controlled gas access, making it a practical and effective safety solution for home kitchens.

V. CONCLUSION

This research paper presents a practical and reliable solution for enhancing gas stove safety through a password-based digital gas stove. By integrating a Arduino with a solenoid valve, keypad, and LCD display, the proposed system effectively restricts unauthorized access, reducing the risk of accidents in domestic kitchens. The system's design is cost-effective, user-friendly, and adaptable, making it a viable solution for households and potentially other applications in commercial settings where controlled access to gas appliances is essential.

Initial testing demonstrated that the system responds accurately to password inputs, successfully controlling the gas flow through the solenoid valve and providing clear status feedback on the LCD display.

This project not only addresses a critical safety issue but also highlights the capability of Arduino, in managing secure access control mechanisms for daily-use appliances. Future work could involve adding features such as remote monitoring or integrating gas leakage detection for a comprehensive kitchen safety solution. The project contributes to the ongoing efforts in home automation and safety, emphasizing the importance of accessible technology in creating safer living environments.

VI. REFERENCES

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